Arista Access Node HEB-ACC-63 QoS Congestion Event Resolution SOP

Node: HEB-ACC-63 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_2132

Description

This Standard Operating Procedure (SOP) outlines the steps to identify and resolve QoS congestion events on Arista Access Node HEB-ACC-63 at the Hebbal Edge site. The occurrence of congestion events, specifically microbursts, interface tail drops, and QoS queue starvation, is attributed to the inherent nature of the Arista EOS (Extensible Operating System) implementation on the Access role. The Access role, as specified, utilizes a shared buffer architecture to manage incoming traffic, which introduces a bottleneck for congested traffic flows. When the buffer is exceeded, the system resorts to tail drop, leading to congestion and subsequent QoS issues. Furthermore, the Access role's reliance on the QoS configuration to manage traffic prioritization can lead to queue starvation, exacerbating congestion. Therefore, this SOP aims to provide a structured approach to identify and rectify these issues.

Pre-requisite Checks

- Ensure remote hands are dispatched to Hebbal Edge for physical site access.

- Verify the status of redundant links to determine if the issue is isolated to a specific link or a broader network condition.

- Review the QoS configuration on the Access node to ensure proper configuration and policy implementation.

Potential Root Causes

Cause	Probability
Microburst or high-traffic event exceeding buffer capacity	High
Inadequate QoS configuration or policy implementation	Medium
Hardware or software issues with the Access node	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface HEB-ACC-63-0/0/1 detail, Expected Output: Show detailed interface statistics, including error counters and packet drops., Description: This command is used to gather detailed statistics on the interface, including error counters and packet drops, to determine if the issue is related to the interface or a broader congestion event.

Command: show qos policy HEB-ACC-63-0/0/1 detail, Expected Output: Show detailed QoS policy configuration and statistics., Description: This command is used to review the QoS policy configuration and
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statistics to determine if the issue is related to proper policy
implementation or configuration.

Command: show processes cpu detail, Expected Output: Show detailed CPU
usage statistics., Description: This command is used to gather
detailed CPU usage statistics to determine if the issue is related to
high CPU utilization or resource contention.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policy configuration, Command: config qos policy HEB-ACC-63-0/0/1, Description: Adjust the QoS policy configuration to ensure proper traffic prioritization and management of congested flows.

Step 2: Step: Verify and adjust interface settings, Command: config interface HEB-ACC-63-0/0/1, Description: Adjust the interface settings to ensure proper interface configuration and management of error counters and packet drops.

Step 3: Step: Verify and adjust CPU resources, Command: config processes cpu, Description: Adjust CPU resources to ensure proper utilization and management of resource contention.

Post-Check Verification

```
Command: show interface HEB-ACC-63-0/0/1 detail, Expected Output:
Verify that error counters and packet drops are within normal ranges.,
Description: Verify that error counters and packet drops are within
normal ranges to ensure that the resolution steps have mitigated the
congestion event.

Command: show qos policy HEB-ACC-63-0/0/1 detail, Expected Output:
Verify that QoS policy configuration and statistics are within normal
ranges., Description: Verify that QoS policy configuration and
statistics are within normal ranges to ensure that the resolution
steps have properly implemented the policy configuration.

Command: show processes cpu detail, Expected Output: Verify that CPU
usage statistics are within normal ranges., Description: Verify that
CPU usage statistics are within normal ranges to ensure that the
resolution steps have properly managed CPU resources.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: config rollback qos policy HEB-ACC-63-0/0/1, Description: Restore the original QoS policy configuration to its previous state to ensure proper rollback procedure in case of failed resolution.

- Step: Restore original interface settings, Command: config rollback interface HEB-ACC-63-0/0/1, Description: Restore the original interface settings to its previous state to ensure proper rollback procedure in case of failed resolution.

- Step: Restore original CPU resources, Command: config rollback processes cpu, Description: Restore the original CPU resources to its previous state to ensure proper rollback procedure in case of failed resolution.